

Project Report on GitOps Workflow using ArgoCD on Kubernetes

Introduction:

GitOps is a modern approach to managing infrastructure and application deployments using Git as the single source of truth. In this project, ArgoCD was used to automate the deployment and synchronization of Kubernetes resources from a GitHub repository. Any changes made to the deployment manifests in GitHub were automatically detected and applied to the Kubernetes cluster by ArgoCD, ensuring consistency and reducing manual deployment efforts.

Abstract:

The objective of this project was to implement a GitOps workflow using ArgoCD and Kubernetes. A local Kubernetes cluster was created using Minikube, and ArgoCD was installed to manage deployments. Kubernetes manifest files were stored in a GitHub repository and configured as the source for ArgoCD. Automatic synchronization was enabled so that any updates pushed to GitHub were automatically deployed to the cluster. The project demonstrated version-controlled deployments, automated synchronization, self-healing capabilities, and efficient application lifecycle management.

Tools Used:

- Git Bash
- GitHub
- Docker Desktop
- Kubernetes (Minikube)
- kubectl
- ArgoCD
- Nginx Sample Application

Steps Involved in Building the Project:

1. Installed Git, Docker Desktop, kubectl, and Minikube on Windows.
2. Started a local Kubernetes cluster using Minikube.
3. Created a dedicated namespace and installed ArgoCD in the cluster.
4. Accessed the ArgoCD web interface and configured authentication.
5. Created Kubernetes deployment and service manifest files for an Nginx application.
6. Created a GitHub repository and pushed the manifest files.
7. Connected the GitHub repository to ArgoCD and created an application.

8. Enabled automatic synchronization, pruning, and self-healing features.
9. Verified successful deployment of the application in Kubernetes.
10. Modified application configurations in GitHub and observed automatic updates through ArgoCD.
11. Captured screenshots and documented the GitOps workflow.

Conclusion:

This project successfully demonstrated the implementation of GitOps using ArgoCD and Kubernetes. By using GitHub as the source of truth, deployments became automated, traceable, and version-controlled. ArgoCD continuously monitored the repository and synchronized the cluster state with the desired configuration. The project provided practical experience with Kubernetes, GitOps principles, continuous deployment, and infrastructure automation, making it a valuable DevOps portfolio project.